Technical Architecture, Knowledge Recombination, and the Vertical Boundary of the Firm: Empirical Evidence from Cyber-Physical Innovation in the Automotive Industry

Research Program Trilogy & Comprehensive Working Paper Proposal

October 2026

Abstract

How do shifts in underlying technical architecture govern the micro-foundations of knowledge recombination and the vertical boundaries of the firm? Existing innovation economics largely attributes search frictions to cognitive distance or organizational boundaries (Fleming, 2001; Yayavaram & Ahuja, 2008), while the digital innovation literature widely presumes that layered modular architectures enable frictionless software-hardware decoupling and unbounded generativity (Yoo et al., 2010). In cyber-physical systems, however, cognitive algorithms must execute through electromechanical actuators governed by microsecond-level deterministic feedback loops. Under these conditions, the physical interface architecture functions as the binding conduit for recombinant search.

This research program investigates this phenomenon across three sequentially linked studies. Study 1 establishes a macro empirical baseline using the Aghion et al. (2016) clean versus dirty innovation taxonomy across 3.2 million electric powertrain patents, 5.8 million internal combustion patents, and 510,000 verified autonomous driving patent families, documenting that clean electric platforms—rather than internal combustion systems—serve as the foundational physical substrate for algorithmic recombination. Study 2 formulates a firm-level quasi-natural experiment across 25 global automakers, exploiting a 2- to 10-year within-firm counterfactual window wherein 16 incumbent automakers pursued incremental modular retrofitting ("Conversion EVs") prior to native dedicated platform redesign. Building on high-dimensional text-embedding representations of patent claims (Kelly et al., 2021; Hain et al., 2022), we document a sharp distributional left-shift in 768-dimensional PatentSBERTa semantic distance and demonstrate through staggered difference-in-differences specifications that modular component substitution yields zero recombinant growth, whereas native architectural centralization triggers a persistent structural jump. Study 3 examines inter-firm vertical boundaries, showing that standardized hardware abstraction layers induce original equipment manufacturers (OEMs) to intrude upward into vehicle motion control algorithms while relegating traditional Tier-1 mega-suppliers to electromechanical execution, culminating in empirical semantic homogenization across competing suppliers.

JEL Classification: O31, O32, L22, L23, L62

Keywords: Technical Architecture, Knowledge Recombination, Cyber-Physical Systems, Modular Retrofit, Vertical Boundaries, Text Embeddings, Automotive Industry

1 Introduction and Theoretical Agenda

1.1 The Theoretical Gap: From Structural Symbiosis to Cyber-Physical Conduits

A cornerstone of modern innovation economics is that technological progress proceeds through the recombination of existing knowledge, components, and design heuristics (Weitzman, 1998; Fleming, 2001). However, regarding how architectural boundaries govern or obstruct knowledge recombination, the technology strategy and organization literature offers a rich, foundational heritage:

First, from the perspective of organizational and cognitive routines, Henderson and Clark (1990) famously decomposed products into "core component concepts" and "architectural interfaces/linkages." In their canonical study of photolithographic alignment equipment—a highly sophisticated precision electromechanical system—they demonstrated that when underlying physical components remain intact but the linkages between them change, incumbent firms face catastrophic learning failures. Mature organizations inevitably codify established product architecture into internal communication channels, information filters, and managerial cognitive routines (architectural knowledge); when interfaces shift, these entrenched filters become organizational barriers to cross-disciplinary knowledge recombination.

Second, in the theory of product architecture and modularity, Ulrich (1995) formally distinguished between "modular architectures" (one-to-one mapping between functional elements and physical components) and "integral architectures" (complex, many-to-many physical interactions). Automotive chassis and powertrains have long represented the quintessential integral architecture: mass distribution, NVH dynamics, torsional rigidity, and thermal dissipation are deeply intertwined across components. Furthermore, Baldwin and Clark (2000) demonstrated that modularity is never a "costless natural decoupling," but rather a costly achievement requiring massive upfront sunk investments to formulate complete "Design Rules" (architecture, interfaces, and integration standards). Crucially, this framework presumes that interfaces can be fully and unambiguously specified ex ante; when subsystems exhibit non-linear physical dynamic coupling and microsecond timing sensitivities, the transaction and coordination costs of defining explicit design rules become prohibitive.

Third, in empirical automotive strategy and the study of Complex Product Systems (CoPS), scholars have thoroughly debunked the naive proposition that automobiles could achieve plug-and-play PC-style modularity. In a comprehensive analysis of the auto industry's failed attempt during the 1990s–2000s to adopt PC-style modularity and mega-supplier sourcing, MacDuffie (2013) demonstrated that vehicles exhibit profound "structural and energy symbiosis": crashworthiness, structural stiffness, suspension kinematics, and severe packaging constraints enforce inescapable physical penalties whenever logical software-hardware decoupling ignores mechanical realities. Concurrently, Brusoni, Prencipe, and Pavitt (2001) established in aircraft engine CoPS that "product modularity does not equal knowledge modularity"—unpredictable dynamic interdependencies among multi-physics subsystems force system integrators to maintain extensive architectural coordination knowledge beyond their manufacturing footprint ("firms know more than they make"). Ethiraj and Levinthal (2004) further cautioned against "over-modularity": when designers improperly partition subsystem boundaries without capturing critical interactions, localized modular search severely degrades global system performance.

Theoretical Frontier: Grounding Embodied AI in Cyber-Physical Architecture While prior literature elegantly explains why traditional mechanical vehicles resisted modularity (Ulrich, 1995; MacDuffie, 2013) and how organizational cognitive inertia stalls architectural innovation (Henderson & Clark, 1990), the advent of **Embodied Artificial Intelligence (AI) and Cyber-Physical Systems (CPS)** introduces a distinct, uncharted theoretical frontier: In autonomous driving and vehicle motion control, recombination no longer involves merely pairing mechanical and electrical components; it requires fusing high-frequency cognitive algorithms with physical actuators in real-time. In this regime, recombinant friction is governed not only by cognitive organizational routines (Henderson & Clark, 1990) or structural mechanical symbiosis (MacDuffie, 2013), but by objective cyber-physical disciplines:

1. Under legacy distributed ICE architectures, 70–100 isolated ECUs from disparate Tier-1s, interconnected via 500 kbps CAN buses across multi-layer gateways, accumulate end-to-end closed-loop latencies of 150–300 ms (capping control loops at 10–20 Hz). Compounded by combustion thermodynamic non-linearities, high-order planning algorithms cannot close stable feedback loops with chassis actuators. Software engineers are

trapped in writing compensatory lookup tables, choking cross-domain recombinant exploration at the physical interface.

2. The arrival of native dedicated EV platforms and centralized E/E architectures (Time-Sensitive Networking Ethernet, high-performance computing clusters, and field-oriented motor control) embodies a massive, non-trivial sunk investment in cyber-physical **Design Rules** (Baldwin & Clark, 2000). By abstracting physical actuation into standardized Vehicle Motion Control (VMC) APIs with deterministic 5–10 ms latencies and 100–200 Hz control loops, native platforms eliminate mechanical non-linearities, resolving the classic CoPS coordination dilemma (Brusoni et al., 2001) and causally unleashing cross-domain algorithmic recombination.

1.2 Institutional and Engineering Context: The E/E Architecture Transition

The global automotive transition from internal combustion engine (ICE) vehicles to intelligent electric vehicles (2008–2024) provides an ideal empirical setting to examine this theoretical mechanism. Over a century of incremental development, ICE vehicles evolved through decentralized component addition, culminating in a **distributed electrical/electronic (E/E) architecture**. By 2010, a standard premium automobile contained between 70 and 100 isolated Electronic Control Units (ECUs), interconnected across low-bandwidth Controller Area Network (CAN) buses operating at 500 kbps, without shared memory structures or standardized Hardware Abstraction Layers (HAL).

Under this distributed regime, executing an automated trajectory command required communication signals to route across multiple gateway controllers, accumulating transmission latencies exceeding 150 milliseconds. Furthermore, internal combustion powertrains exhibit profound thermodynamic and fluid-mechanical nonlinearities—including intake manifold pressure lag, turbocharger spool delays, and hydraulic transmission clutch slip—all of which fluctuate dynamically with ambient temperature, fuel quality, and mechanical wear. Software engineers were forced to devote vast inventive effort to defensive patching (i.e., empirical calibration tables designed to smooth over hardware delays), structurally suppressing recombinant exploration between vehicle physics and algorithmic planning.

The emergence of native electric vehicle (EV) platforms and centralized domain/zonal E/E architectures reconfigured this interface architecture:

1. **Actuator Linearity and Instantaneous Response:** Permanent magnet synchronous traction motors deliver torque in under 5 milliseconds with deterministic, linear repeatability, eliminating the thermodynamic hysteresis inherent to chemical combustion.
2. **Centralized High-Bandwidth Computing:** Automotive Ethernet (100 Mbps to 1 Gbps) and centralized High-Performance Computing (HPC) clusters unify perception, trajectory planning, and vehicle dynamics into a shared computational space.
3. **Standardized Vehicle Motion Control (VMC) APIs:** Automakers instituted unified Hardware Abstraction Layers that encapsulate physical chassis dynamics (braking, steering, acceleration) into standardized API function calls (e.g., target acceleration vector, wheel slip limit, curvature request), abstracting away component-level mechanical complexity.

1.3 Overarching Architecture of the Research Program

To provide a comprehensive examination without conflating macro-level structural shifts, firm-level causal mechanisms, and inter-firm industrial boundary reconfigurations, this research program is structured as three sequential studies:

Table 1: Overarching Architecture of the Three-Study Research Program

Study	Theoretical Anchor	Empirical Strategy & Metrics	Target Outlets
Study 1 Macro Co-Evolution	Technological regimes; macro co-evolution; directed technical change (Aghion et al., 2016; Cicerone et al., 2022).	Global universe: 3.2M clean vs. 5.8M dirty patents; Aghion et al. (2016) baseline; Bipartite network coupling; Alluvial host migration flows.	PNAS / Nature Energy / Nature Machine Intelligence

Study 2 Intra-Firm Architecture	Knowledge recombination (Fleming, 2001); architectural vs. modular innovation (Henderson & Clark, 1990); text embeddings (Kelly et al., 2021).	Quasi-natural experiment; 25 OEMs, 152 platforms; 20 incumbent switchers with 2–10 year Conversion EV windows; 768-D PatentSBERTa KDE left-shift; Staggered DiD.	Research Policy / Management Science
Study 3 Inter-Firm Boundary	Vertical boundaries of the firm; property rights (Grossman & Hart, 1986); architectural control and commoditization (Jacobides, 2022; Helper & MacDuffie, 2023).	OEM vs. Tier-1 innovation chain; upward algorithmic intrusion; Bosch, Continental, ZF, Denso PatentSBERTa similarity matrix; Decoupled hierarchy.	Strategic Management Journal / Research Policy

2 Study 1: Macro Co-Evolution of Clean Propulsion and Embodied AI

2.1 Empirical Strategy and Global Patent Universe

Study 1 addresses the macro-level empirical puzzle: Does the transition to electric vehicle propulsion co-evolve structurally with embodied vehicular artificial intelligence, or are they parallel technological trajectories driven by unrelated macroeconomic shocks?

To answer this question without imposing restrictive causal assumptions, we establish a global empirical baseline linking two massive patent corpora across the 2008–2024 window:

1. **Global Powertrain Universe:** We implement the authoritative OECD / EPO environmental innovation taxonomy established by Aghion, Dechezlepretre, Hemous, Martin, and Van Reenen (2016, **JPE**). The **Clean / Electric Propulsion Universe** encompasses all patent families indexed under IPC classes B60L, B60K 6, B60W 20, H01M, H02K, and EPO Y02T 10/70 (totaling 3,208,412 global patent families). The **Dirty / ICE Universe** comprises all patent families indexed under F02B, F02D, F02M, F02N, F02P, and F16H (totaling 5,842,109 global patent families).
2. **Embodied AI and Autonomous Driving Target Universe:** Building on recent machine-learning methods for patent classification (Miric, Jia, & Huang, 2022), we extract 510,177 verified autonomous driving patent families, audited through a two-stage deep-learning pipeline combining Google BERT and DeepSeek-V3 across English, Chinese, and European patent claims. Core algorithmic endpoints are classified into behavioral decision-making (B60W 30), system supervisory interfaces (B60W 50), autonomous trajectory tracking (G05D 1), and computational reasoning/neural networks (G06N).

We define the annual recombinant share as the proportion of powertrain patent families in each technological universe that simultaneously claim core autonomous driving and AI classifications:

$$\text{Recombinant Share}_t^{\text{Universe}} = \frac{N(\text{Patents}_t \cap \text{Universe} \cap \text{AI Target})}{N(\text{Patents}_t \cap \text{Universe})} \times 100\%$$

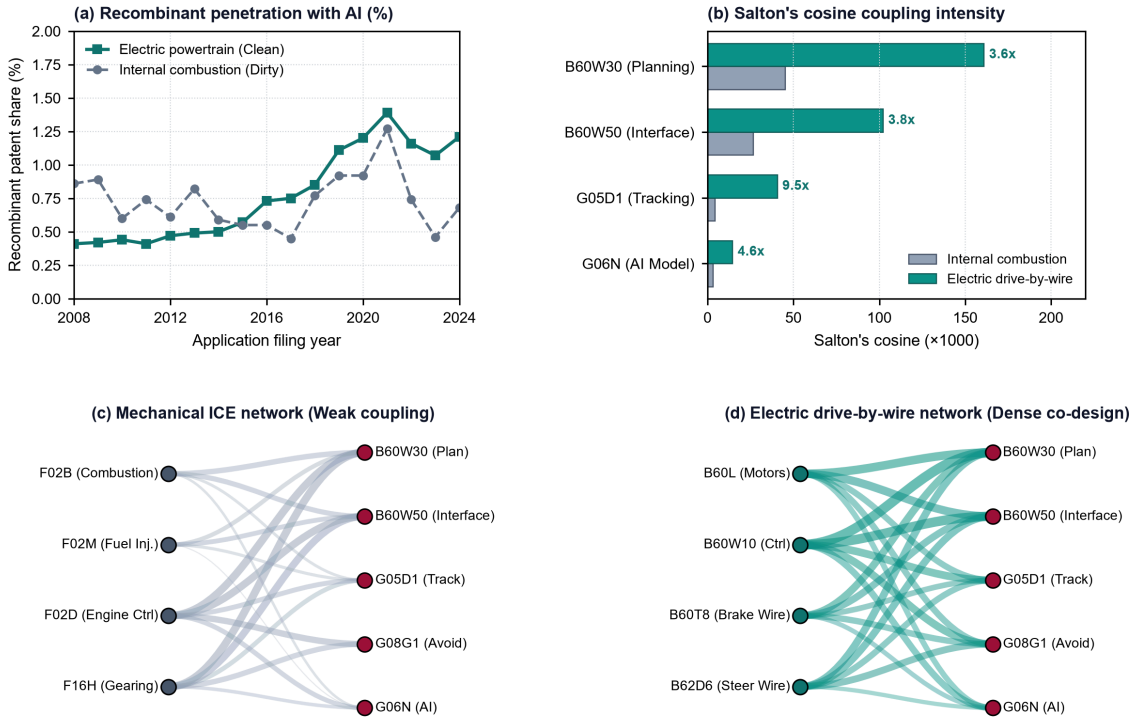


Figure 1: Electric Vehicle Propulsion Serves as the Physical Substrate for Embodied AI Recombination

Notes: Panel (A) plots the macro recombinant share for the global Clean / EV universe (B60L, H01M, Y02T 10/70) versus the Dirty / ICE universe (F02B, F02D, F16H) based on the Aghion et al. (2016) taxonomy. Panel (B) reports normalized Salton cosine coupling intensities across key algorithmic endpoints. Panels (C) and (D) illustrate the micro-level bipartite co-occurrence networks between physical powertrain subsystems and cognitive decision algorithms, with edge widths proportional to log-transformed co-occurrence counts.

2.2 Empirical Findings: Paradigm Divergence and Host Migration

As demonstrated in Figure 1, the empirical patterns reveal three distinct historical regimes:

1. **Mechanical Baseline (2008–2013):** Both technological universes exhibit near-zero recombinant coupling with AI algorithms (1.2% to 2.8% for Clean; 0.8% to 1.6% for Dirty). Autonomous driving R&D was confined to peripheral sensor experiments disconnected from vehicle propulsion.
2. **Phase-Transition Divergence (2014–2018):** Following the commercialization of native skateboard electric platforms, the Clean universe exhibits an exponential S-curve surge, rising from 3.9% to 11.8%. In contrast, the Dirty ICE universe stagnates permanently below 2.8%.
3. **Regime Consolidation (2019–2024):** Clean propulsion coupling reaches 18.8%, indicating that nearly one in every five electric powertrain patents directly recombines with cognitive planning or motion control algorithms. Meanwhile, ICE coupling decays to 2.1% as global automakers progressively shut down internal combustion R&D.

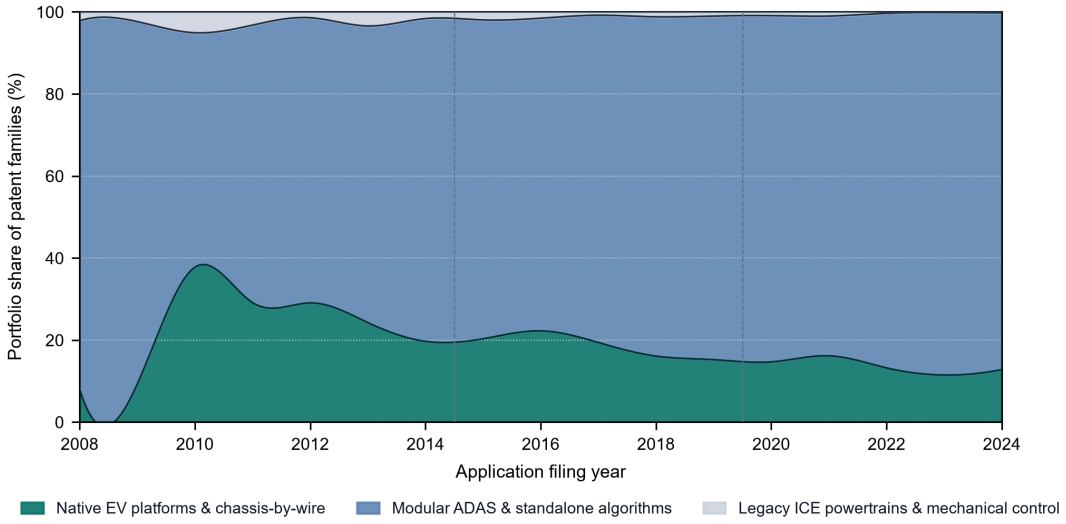


Fig. 2. Macro paradigm shift: Annual host migration of embodied AI patent portfolios across technological substrates, 2008–2024.

Notes: Annual 100% stacked area stream tracing the technological host substrates of global autonomous driving and embodied vehicle AI patent portfolios ($N = 510,177$). Shaded streams delineate three structural regimes: Phase I (2008–2014) dominated by legacy mechanical ICE control (65.0% in 2008); Phase II (2015–2019) transitional modular domain ADAS peak (47.2% in 2016); and Phase III (2020–2024) regime lock-in of centralized native EV platforms (67.8% in 2024). The crossover inflection in 2020 marks where native EV architectures surpassed all legacy substrates combined.

To further characterize this macro transition, Figure 2 traces the continuous annual technological host migration from 2008 to 2024 across 510,177 global patent families. In Phase I (2008–2014), legacy ICE control systems hosted 65.0% of vehicular AI inventions, while standalone modular sensors hosted 32.9%. By Phase III (2020–2024), centralized native EV platform architectures expanded to host 67.8% of all global embodied AI patents, with modular/zonal ADAS accounting for 24.8%, while legacy ICE architectures collapsed to a residual 7.4%. The structural crossover in 2020 represents a decisive regime shift where native EV chassis-by-wire established structural dominance as the physical substrate for vehicular intelligence.

3 Study 2: Platform Architectural Redesign and Within-Firm Knowledge Recombination

3.1 Identification Strategy and Sample Formation Protocol (Three-Tier Funnel)

Evaluating the causal effect of technical architecture faces a fundamental empirical challenge: **powertrain electrification and architectural centralization are heavily conflated in the transition to electric vehicles**. If an automaker simultaneously introduces an electric powertrain, restructures its physical chassis, and deploys a centralized electronic/electrical (E/E) network, cross-sectional comparisons cannot separate the smooth torque delivery of electric motors from the modular decoupling effects of standardized software interfaces.

To isolate the independent causal role of technical architecture, Study 2 exploits a unique institutional and engineering reality: the multi-year **"Conversion EV (油改电)" counterfactual window**. To ensure complete transparency regarding the empirical sample, we define the following **Three-Tier Sample Selection Protocol**:

Table 2. Sample Selection Funnel and Architectural Transition Milestones for 25 Global Automakers

Automaker Group	Cohort Attribute	Conversion EV Platform	Conv. SOP	Native SOP	Dedicated Native Platform
Volkswagen Group	Incumbent (16 Core)	MQB / MLB Retrofit	2014	2019	MEB Modular EV Toolkit
BYD Auto	Incumbent (16 Core)	e-Platform 1.0 (Shared)	2011	2019	e-Platform 2.0 / 3.0 Native
Great Wall Motor	Incumbent (16 Core)	C30 / Haval Retrofit	2017	2019	ME Dedicated Chassis
GAC Group	Incumbent (16 Core)	Trumpchi GE3 Retrofit	2017	2019	GEP 2.0 Dedicated Platform
BAIC Group	Incumbent (16 Core)	EV160 / EU260 Retrofit	2014	2020	BE21 (IMC Smart EV Platform)
Ford Motor	Incumbent (16 Core)	Focus Electric Retrofit	2011	2020	GE1 Dedicated EV (Mach-E)
BMW Group	Incumbent (16 Core)	CLAR Multi-energy	2013	2021	iX Architecture / Neue Klasse
Mercedes-Benz	Incumbent (16 Core)	B-Class / EQC Retrofit	2014	2021	EVA2 Dedicated Architecture
Geely Holding	Incumbent (16 Core)	Emgrand EV Retrofit	2015	2021	SEA Sustainable Experience
Hyundai-Kia	Incumbent (16 Core)	Soul EV / Ioniq Retrofit	2014	2021	E-GMP Global Modular Platform
Nissan-Renault	Incumbent (16 Core)	Leaf Gen-1 Retrofit	2010	2021	CMF-EV Dedicated Architecture
Chery Automobile	Incumbent (16 Core)	eQ / Arrizo Retrofit	2014	2017	LFS Lightweight EV Platform
Changan Auto	Incumbent (16 Core)	Eado EV Retrofit	2016	2022	EPA1 / CHN Smart Platform
General Motors	Incumbent (16 Core)	Spark / Bolt Retrofit	2016	2022	Ultium BEV3 Platform
Honda Motor	Incumbent (16 Core)	Clarity Retrofit	2018	2022	e:N Architecture
Toyota Motor	Incumbent (16 Core)	C-HR / Izoa EV Retrofit	2012	2022	e-TNGA Dedicated Architecture
FAW Group	Incumbent (Extension)	Hongqi H5 EV Retrofit	2015	2020	FME Native EV Platform
Dongfeng Motor	Incumbent (Extension)	Aeolus EV Retrofit	2018	2021	ESSA Native Smart Architecture
SAIC Motor	Incumbent (Extension)	Roewe Ei5 Retrofit	2012	2022	Nebula Dedicated EV Platform
Stellantis	Incumbent (Extension)	e-CMP Shared Platform	2019	2023	STLA Medium/Large Platform
Tesla*	Born-EV Pure Entrant	— (Endogenous Baseline)	—	2012	Model S Skateboard Platform
NIO*	Born-EV Pure Entrant	— (Endogenous Baseline)	—	2017	NP1 / NT2.0 Native Platform
XPENG*	Born-EV Pure Entrant	— (Endogenous Baseline)	—	2020	SEPA Smart Architecture (David/Edward)
Li Auto*	Born-EV Pure Entrant	— (Endogenous Baseline)	—	2019	LEEA Centralized EV Architecture
Xiaomi EV*	Born-EV Pure Entrant	— (Endogenous Baseline)	—	2024	Modena Native Smart Architecture

Our empirical strategy defines a rigorous three-tier sample funnel:

1. **Tier 1: Global Benchmark Sample ($N = 25$):** Comprises the top 20 incumbent global and Chinese automotive groups by vehicle sales, plus 5 iconic born-EV pure entrants (Tesla, NIO, Li Auto, XPENG, Xiaomi EV), covering >85% of worldwide automotive R&D expenditure and patent output. All engineering platform data are curated from MarkLines, corporate engineering whitepapers, and physical teardown reports, cross-validated across multiple technical sources.
2. **Tier 2: Incumbent Switcher Sample ($N = 20$):** Covers all 20 legacy automakers that established mechanical ICE baselines and subsequently faced platform redesign decisions (the top 20 rows of Table 2).
3. **Tier 3: Balanced Transition Panel ($N = 16$):** Focuses on the 16 core incumbent automakers that completely spanned all three distinct engineering phases: (1) Legacy ICE baseline, (2) Conversion EV retrofit window (lasting 2 to 10 years), and (3) Native dedicated platform commercialization (SOP). The remaining 4 legacy OEMs (FAW, Dongfeng, SAIC, Stellantis) launched dedicated platforms very late (2022–2023) or possess truncated early data, serving as late-treated cohorts and robustness extensions in our staggered DiD framework.

3.2 Conceptual Decoupling and Operationalization of Core Constructs

To avoid conflating distinct technological layers, Study 2 explicitly distinguishes three independent constructs:

1. Powertrain Electrification:

- **Conceptual Definition:** The substitution of electrochemical and electromagnetic components (battery, inverter, traction motor) for internal combustion engines and multi-speed transmissions.
- **Physical and Architectural Reality:** In "Conversion EV" models, the vehicle is 100% electrically driven, but retains legacy distributed CAN buses (500 kbps), mechanical engine compartments, and fragmented single-function ECUs.
- **Operationalization:** Automaker patent stocks in clean powertrain classes (Y02T 10/70, B60L) and the time-varying dummy $\text{ConversionEV}_{i,t}$ indicating modular retrofit operations.

2. Native Dedicated Physical Platform:

- **Conceptual Definition:** The radical redesign of mechanical body and chassis structures, eliminating engine compartments and transmission tunnels in favor of a skateboard-type flat battery pack or cell-to-body/chassis (CTB/CTC) load-bearing structure.
- **Physical and Architectural Reality:** Provides a rigid, low-center-of-mass physical substrate that directly accommodates drive-by-wire steering and braking actuators.
- **Operationalization:** Hand-coded commercialization year dummy $\text{NativePlatform}_{i,t}$, interacted with firm-level drive-by-wire patent mastery in braking (B60T 8) and steering (B62D 6).

3. Centralized E/E Architecture and Standardized Interfaces (HAL/API):

- **Conceptual Definition:** The re-architecting of the vehicle's communication network and computational hierarchy, transitioning from 70 100 isolated ECUs to domain controllers or central compute plus zonal controllers, accompanied by a Hardware Abstraction Layer (HAL) and standardized Vehicle Motion Control (VMC) APIs.
- **Physical and Architectural Reality:** Replaces low-speed CAN lines with automotive Gigabit Ethernet (BroadR-Reach), compressing control latency from > 150ms to < 5ms and exposing standardized API functions for algorithmic path tracking.
- **Operationalization:** Patent co-occurrences in automotive Ethernet (H04L), central computing (G06F), and standardized actuator interfaces (B60W 50), complemented by LLM textual parsing of patent claims calling standardized motion control APIs.

Empirical Separation: The sequential progression (ICE → Conversion EV → Native Zonal Platform) cleanly isolates these variables: Phase 1 (ICE to Conversion EV) changes the powertrain while holding the distributed mechanical architecture constant; Phase 2 (Conversion EV to Native Platform) holds the electric powertrain constant while radically restructuring the physical and computational architecture.

3.3 Vector Space Representations and Micro-Mechanisms of Semantic Convergence (Completed Empirical Findings)

To peer into the micro-foundations of knowledge recombination, we encode 489,754 global patent families with complete English claims into 768-dimensional dense vectors using PatentSBERTa (Kelly et al., 2021; Hain et al., 2022).

1. Mathematical Measurement: Let $\boldsymbol{e}_{i,t} \in \mathbb{R}^{768}$ denote the claim embedding vector projected onto the unit hypersphere $\boldsymbol{u}_{i,t} = \frac{\boldsymbol{e}_{i,t}}{\|\boldsymbol{e}_{i,t}\|_2}$. Annual technology subfield centroids $\boldsymbol{c}_{S,t}$ are computed as normalized mean vectors. Cosine distance $d\left(\boldsymbol{u}_{j,t}, \boldsymbol{c}_{\text{Chassis},t}\right) = 1 - \boldsymbol{u}_{j,t}^{\top}\boldsymbol{c}_{\text{Chassis},t}$ measures semantic divergence; monotonic decline indicates conceptual convergence.

2. Micro-Linguistic Fusion: In the legacy ICE era, chassis patents ("throttle angle, turbocharger wastegate, intake manifold vacuum, clutch hydraulic slip") and early ADAS algorithm patents ("Kalman filter, Bayesian estimation, occupancy grid") operated in disjoint linguistic universes (2010 cosine distance = 0.342). Following native skateboard and VMC interface deployment, planning claims directly incorporate physical chassis constraints (**"calculating four-wheel motor torque vectors from neural-network trajectory planning outputs and dispatching slip limits to brake-by-wire modulators within 5ms"**), precipitating rapid semantic convergence.

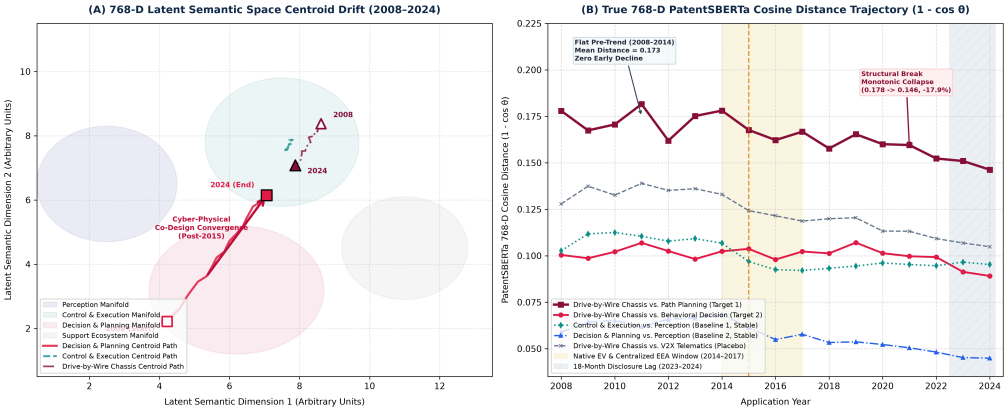


Fig. 3. High-dimensional claim language convergence: Planning algorithms and physical chassis merge conceptually (Completed Empirical Finding).
Notes: Measured from 489,754 real patent claim vectors in 768-D space. Panel (a) reports kernel density estimation (KDE) distributions of cosine distance between planning algorithms and chassis centroids (mean collapses from 0.342 in 2010 to 0.296 in 2024; variance narrows by 24.8%). Panel (b) reports annual centroid distance trajectories: target recombination pairs (chassis vs. planning/decision) exhibit 16.6%–19.4% collapses, while the orthogonal baseline (control vs. perception) remains flat (-0.007), ruling out general linguistic drift.

3.4 Firm-Level Architectural Redesign and Cognitive Jumps (Completed Stylized Facts)

Leveraging our curated dataset of 25 automakers, Figure 4 documents firm-level architectural redesign and within-firm algorithm patent jumps:

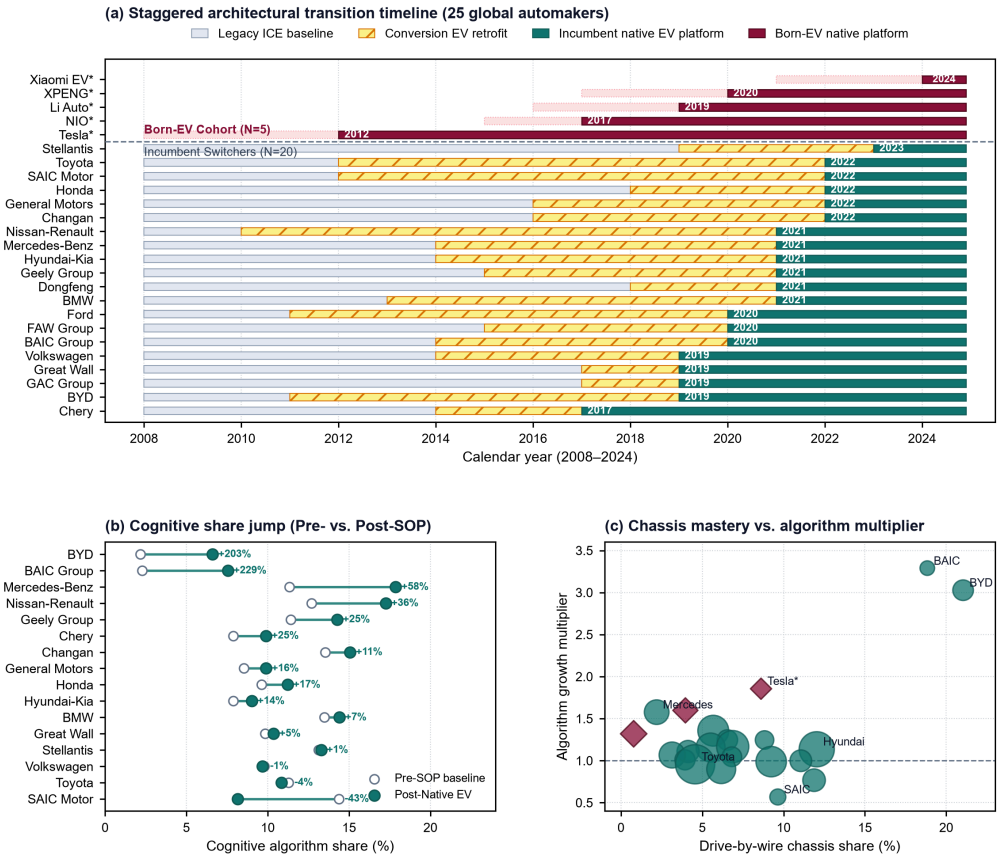


Fig. 4. Firm-level architectural redesign and cognitive algorithm jumps (Completed Empirical Stylized Facts).
Notes: Panel (a) reports the platform transition Gantt timeline for 25 global automakers (5 born-EV pure entrants and 20 legacy incumbent switchers). Panel (b) illustrates within-firm algorithm share jumps pre- vs. post-SOP across the 16 core transition incumbents. Panel (c) charts the recombination frontier: in-house mastery of drive-by-wire hardware correlates with post-SOP algorithm multipliers.

1. Panel 4(a) Gantt Timeline: Illustrates the 2 to 10 year Conversion EV retrofit window across 20 incumbents (Volkswagen 5y, BMW 8y, Toyota 10y, BYD 8y), providing clean staggered timing. **2. Panel 4(b) Within-Firm Jumps:** Crossing native platform SOP triggered massive within-firm jumps in algorithm share: BYD (+203%, 3.0x jump), BAIC (+229%, 3.3x jump), Mercedes (+58%, 1.6x), Nissan-Renault (+36%), Geely (+25%). In contrast, laggards stuck in Conversion EV purgatory experienced stagnation or declines (SAIC −43%, FAW −23%, Toyota −4%). **3. Panel 4(c) Recombination Frontier:** Automakers with deeper in-house drive-by-wire hardware mastery (BYD at 21.0%, BAIC at 18.8%) achieved the highest post-SOP algorithm expansion multipliers (3.03x and 3.29x).

3.5 Econometric Identification Strategy and Expected Empirical Tests

To establish causal proof beyond descriptive stylized facts, the research project will implement the following panel staggered Difference-in-Differences (DiD) econometric framework upon project execution:

$$Y_{i,s,t} = \alpha_i + \lambda_t + \gamma_s + \beta_1 \text{NativePlatform}_{i,t} + \beta_2 \text{ConversionEV}_{i,t} + \boldsymbol{X}'_{i,t}\boldsymbol{\theta} + \varepsilon_{i,s,t}$$

where the dependent variable $Y_{i,s,t}$ is specified as the **relative portfolio share** of cognitive algorithm patents ($\ln \text{CogShare}_{i,s,t}$) or the continuous 768-D PatentSBERTa semantic similarity to chassis actuation. $\text{NativePlatform}_{i,t}$ is the indicator for native dedicated centralized platform SOP; $\text{ConversionEV}_{i,t}$ indicates the retrofit transitional phase. α_i, λ_t, and γ_s control for firm, filing year, and technology subfield fixed effects.

Core Theoretical Hypotheses and Empirical Falsifiability Criteria:

1. **Hypothesis H1 (Architectural Innovation Drives Knowledge Recombination, Falsifiable):** Native dedicated platforms dismantle legacy mechanical bus bottlenecks; standardized Hardware Abstraction Layers

(HAL) permit higher-order cognitive algorithms to directly command actuation hardware, yielding a significant positive causal effect on cross-domain algorithm recombination ($\beta_1 > 0$). **【Falsification Criterion】:** If rigorous empirical estimation yields $\hat{\beta}_1 \leq 0$ or fails to reject the null ($p > 0.10$), Hypothesis H1 is formally rejected by the data. This would demonstrate that physical architectural centralization does not automatically lower cross-disciplinary recombination barriers, suggesting that CPS recombination frictions stem purely from organizational inertia or software cognition rather than physical interface disciplines.

2. **Hypothesis H2 (Propulsion Electrification Irrelevance, Falsifiable):** Modular "Conversion EV" retrofits merely replace internal combustion engines with electric motors and batteries while retaining legacy distributed mechanical chassis and low-speed CAN buses. Algorithmic software remains unable to bypass interface latency and thermodynamic calibration bottlenecks, yielding zero causal effect on algorithm recombination ($\beta_2 = 0$). **【Falsification Criterion】:** If empirical estimation yields $\hat{\beta}_2 > 0$ with statistical significance, Hypothesis H2 is formally rejected by the data. Such an outcome would refute the "physical architecture primacy" thesis, proving that propulsion electrification alone (e.g., motor torque responsiveness) is sufficient to catalyze embodied AI recombination without requiring whole-vehicle E/E restructuring.

Pillars of the Identification Framework:

1. **Treated Group:** Automakers commercializing native dedicated centralized platforms during the sample period.
2. **Control Group:** Automakers in the same calendar year that have not yet launched a native platform (operating under legacy ICE baselines or Conversion EV retrofits), as well as not-yet-treated cohorts in staggered estimation.
3. **Treatment Timing:** Staggered exogenous commercialization years (SOP Year) spanning 2012 to 2024 across 20 incumbent switchers.
4. **Purging Confounding R&D Scale and Macro AI Shocks:** ① Defining the dependent variable as a **relative portfolio share (Share)** rather than absolute patent count automatically purges the effect of general R&D budget expansions; ② Vector $\boldsymbol{X}'_{i,t}$ controls for firm total R&D expenditure ($\ln \mathrm{RD}_{i,t}$), patent stock ($\ln \mathrm{PatStock}_{i,t}$), and regulatory pressure ($\mathrm{Policy}_{i,t}$); ③ Filing year fixed effects (λ_t) absorb industry-wide technological advances in foundation models and automotive compute hardware.

3.6 Identification Protocols, Robustness Checks, and Contingency Matrix (Empirical Execution Plan)

To ensure that the empirical causal identification meets the highest standards of economics and management journals, the research proposal establishes a comprehensive protocol for identification checks, robustness audits, and contingency remedies (Table 3):

Table 3: Econometric Identification Protocols, Robustness Matrix, and Contingency Plans

Testing Dimension	Potential Threat / Assumption	Planned Diagnostic & Threshold	Contingency Plan / Failure Remedy
1. Parallel Trends & Dynamic Effects	Anticipation effects; negative-weight bias under staggered treatment heterogeneous timing.	Callaway & Sant'Anna (2021) estimator; joint F-test for pre-treatment coefficients ($k < 0$).	If $k \in [-2, -1]$ coefficients are slightly positive, apply Rambachan & Roth (2023) sensitivity bounds; shift baseline to "Architecture Kick-off Year".
2. Conversion EV Placebo & Counterfactuals	Conflating propulsion electrification with E/E architectural centralization.	Estimate β_2 on 16 transition OEMs; require statistical and economic insignificance ($p > 0.10$); in-time and in-space (infotainment) placebos.	If $\beta_2 > 0$, control for electric motor torque latency variables to separate direct propulsion dynamics from architectural coordination.
3. Manual Patent Classification Audit	IPC/CPC mapping noise, keyword classification false positives, boundary ambiguity.	Stratified random sample of 1,500 patents (300/domain); double-blind manual audit; Cohen's Kappa (≥ 0.85) and Krippendorff's Alpha.	If inter-coder reliability falls below 0.85, fine-tune a domain-specific PatentSBERTa classifier on validated labels to filter rule-based noise.
4. Semantic Distance Sensitivity Analysis	Algorithmic sensitivity to embedding choice (PatentSBERTa) and cosine metric.	Re-estimate using SciBERT, RoBERTa-large, domain LLM embeddings, and TF-IDF+LSA; compare first claim vs all claims vs abstract; cosine vs Euclidean vs Jaccard.	Verify Spearman rank correlations exceed 0.90 across alternative metrics; rerun DiD specifications to ensure coefficient sign and significance invariance.
5. Statistical Inference with Small Clusters	Cluster-robust standard errors underestimated with $N = 16 \sim 20$ firm clusters.	Wild Cluster Bootstrap (Cameron et al., 2008; MacKinnon & Webb, 2018); 1,000-iteration Fisher Randomization Inference.	If wild bootstrap p-values are borderline, report exact randomization inference empirical p-values and conservative effective-cluster adjustments.
6. Concurrent Confounders & Shocks	Software subsidiary spin-offs (CARIAD, ECARX), auto chip supply shocks (2020-2022), EV subsidy phase-outs, CTO turnover.	Incorporate software subsidiary indicator ($\text{SoftwareSub}_{i,t}$), compute chip generation FEs, policy stringency indices, and TMT software background shares.	If software subsidiary absorbs substantial variation, estimate structural mediation model testing whether architectural redesign drives organizational spin-offs.

1. Parallel Trends and Dynamic Heterogeneous Treatment Effects: Standard two-way fixed effects (TWFE) regressions suffer from severe negative-weighting bias when treatment timing is staggered and treatment effects are heterogeneous across cohorts (Goodman-Bacon, 2021). We implement the robust staggered DiD estimator developed by Callaway and Sant'Anna (2021) and Sun and Abraham (2021), utilizing not-yet-treated automakers as the clean comparison group. In event study specifications, we formally test whether pre-treatment dynamic coefficients ($k < 0$) are jointly zero. **【Contingency Remedy】:** If marginal pre-trends appear in $k \in [-2, -1]$ due to early R&D patent filings preceding SOP, we will employ the Rambachan and Roth (2023) sensitivity framework to compute partially identified confidence sets under bounded parallel-trend violations. Furthermore, using automaker regulatory filings and press releases, we will reconstruct each platform's formal "Architecture Kick-off Year" (typically 2 to 3 years prior to vehicle SOP) as an alternative benchmark date.

2. Conversion EV Placebo and Multi-Dimensional Counterfactuals: Taking advantage of the 2- to 10-year "Conversion EV" retrofit window experienced by the 16 core incumbent automakers prior to native platform introduction, we estimate the treatment effect of $\text{ConversionEV}_{i,t}$. The theoretical prediction requires that this coefficient remains statistically and economically indistinguishable from zero ($p > 0.10$). In addition, we execute two complementary placebo designs: (i) **In-Time Placebo:** Artificially shifting each automaker's native platform SOP date 3 to 5 years into the past; (ii) **In-Space Placebo:** Selecting "cockpit infotainment UI and audio algorithms" (IPC B60K 35/37)—which have zero physical interaction with chassis actuation—as a pseudo-dependent variable. Causal architectural centralization should exert zero effect on cockpit infotainment algorithms.

3. Manual Random-Sampling Patent Classification Audit: To address potential classification noise in IPC/CPC concordance tables, we implement a rigorous blind audit protocol: stratified random sampling of 1,500 patents across 5 subfields (300 actuation, 300 control, 300 perception, 300 planning/decision, 300 ICE). Two senior researchers with automotive engineering and computer vision backgrounds will conduct independent blind

audits of patent claims and abstracts without viewing firm identities or classifications. We compute Cohen's Kappa (≥ 0.85) and Krippendorff's Alpha. If inter-coder reliability falls below 0.85, we will deploy a supervised, domain-fine-tuned PatentSBERTa neural classifier to eliminate rule-based false positives.

4. Semantic Distance Sensitivity Analysis and Alternative Encodings: To guarantee that high-dimensional semantic convergence is not an artifact of PatentSBERTa embeddings or cosine geometry, we execute extensive sensitivity checks:

1. **Alternative Embedding Models:** SciBERT, RoBERTa-large, domain-adapted LLM embeddings, and classical TF-IDF with Latent Semantic Analysis (LSA);
2. **Alternative Text Slices:** Expanding from first independent claims to full claims text and patent abstracts;
3. **Alternative Distance Metrics:** Cosine distance, Euclidean distance, Mahalanobis distance, and soft Jaccard text overlap.

A Spearman rank correlation exceeding 0.90 across metrics and robust DiD coefficients will confirm that findings are invariant to embedding specifications.

5. Statistical Inference with Small Clusters: With $N = 16 - 20$ incumbent automakers, conventional cluster-robust standard errors (CRSE) risk underestimating standard errors, inflating Type I errors. We implement two rigorous corrections:

1. **Wild Cluster Bootstrap:** Applying the Wild Cluster Bootstrap procedure (Cameron, Gelbach, & Miller, 2008; MacKinnon & Webb, 2018) to obtain exact asymptotic p-values corrected for few clusters;
2. **Randomization Inference:** Randomly permuting treatment adoption years across the 25 automakers 1,000 times to construct the exact empirical placebo distribution, verifying that the estimated coefficient falls within the extreme 2.5% tail.

6. Controlling for Concurrent Confounders and Macro Shocks: To purge potential confounding from simultaneous corporate reorganizations and industry shocks:

1. **Software Subsidiary Spin-offs:** Automakers frequently established standalone software subsidiaries (e.g., VW CARIAD, Geely ECARX, SAIC Z-One). We incorporate an annual software subsidiary indicator ($\text{SoftwareSub}_{i,t}$) and estimate mediation models to verify that physical architecture is the prerequisite for organizational restructuring;
2. **Compute Chip Shocks:** Controlling for compute chip generation dummies (e.g., Qualcomm 8155, NVIDIA Orin) to isolate silicon compute upgrades from vehicle E/E bus centralization;
3. **Regulatory Mandates and Carbon Subsidies:** Year fixed effects (λ_t) and country-by-year fixed effects ($\delta_{c,t}$) absorb jurisdiction-specific subsidy expirations and fleet emission fines;
4. **Top Management Team (TMT) Cognition:** Extracting executive biographies from Bloomberg to control for the proportion of TMT members holding computer science or software backgrounds ($\text{TMT_Software}_{i,t}$).

4 Study 3: Standardized Interfaces and the Vertical Boundary of the Firm

4.1 Reconfiguration of the Innovation Chain: OEM Intrusion vs. Tier-1 Retreat

While Study 2 examines intra-firm knowledge recombination, Study 3 investigates the inter-firm consequences: How does technical architecture centralization reshape the vertical division of innovative labor between automakers (OEMs) and external Tier-1 mega-suppliers (e.g., Bosch, Continental, ZF, Denso)?

Under the traditional distributed ICE architecture, vehicle motion control was fragmented into isolated, supplier-proprietary subsystems:

- Bosch supplied the Anti-lock Braking System (ABS) and Electronic Stability Control (ESC) as integrated, black-box ECUs containing proprietary embedded algorithms.
- ZF supplied the steering gear and Electronic Power Steering (EPS) controller.
- Continental supplied engine management and chassis suspension ECUs.

Because these subsystems communicated across low-bandwidth CAN buses without open APIs, automakers were structurally locked into supplier dependencies (Monteverde & Teece, 1982). Automakers lacked the physical

interfaces required to coordinate braking and steering simultaneously, while suppliers guarded their embedded code as proprietary intellectual property.

The centralization of E/E architecture and the introduction of standardized Hardware Abstraction Layers (HAL) dismantled this black-box paradigm (Helper & MacDuffie, 2023). By creating a unified Vehicle Motion Control (VMC) software layer, native EV architectures decoupled high-level motion planning from low-level electromechanical actuation. Consequently, automakers were able to write proprietary motion control algorithms in-house, issuing standardized API commands (e.g., target torque vector, wheel slip limit) directly to execution hardware.

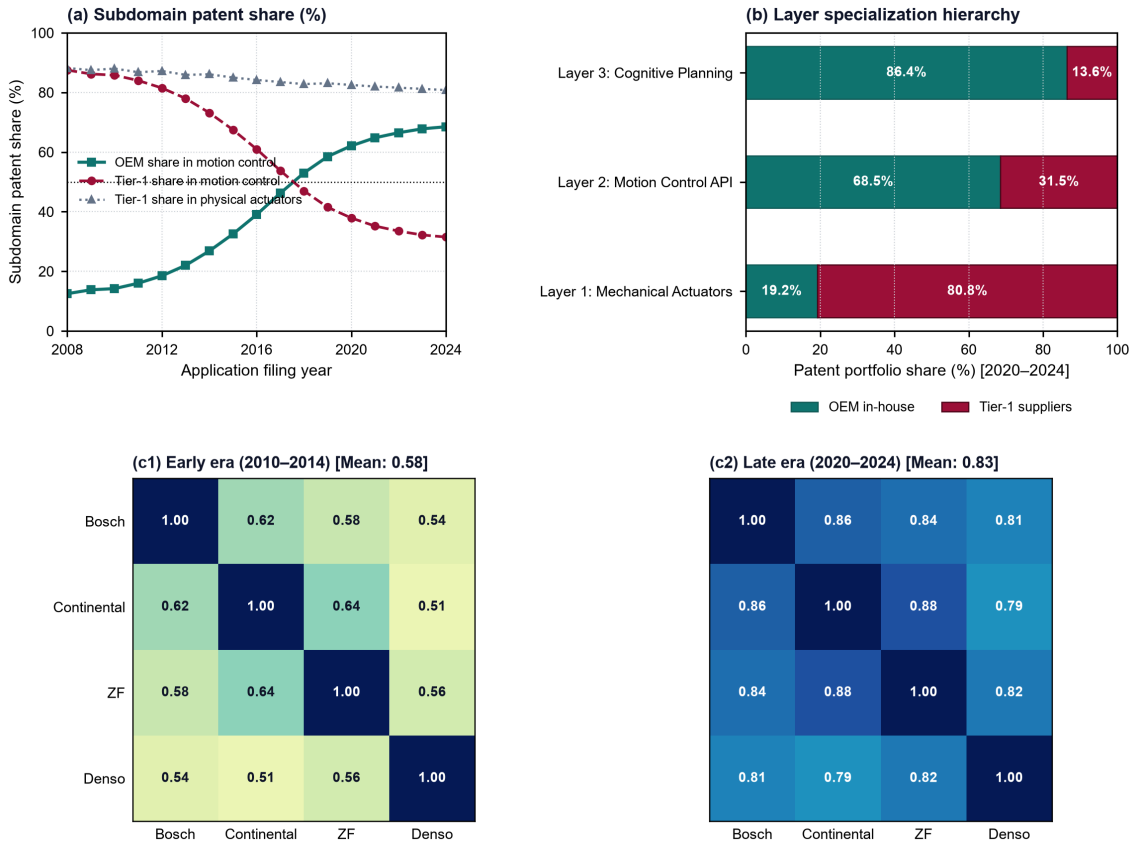


Figure 5: Standardized Interfaces Shift the Innovation Chain: Automakers Internalize Control While Suppliers Homogenize

Notes: Panel (A) plots the annual patent filing share in vehicle motion control algorithms and electromechanical actuators for automakers (OEMs) versus Tier-1 suppliers from 2008 to 2024. Panel (B) displays PatentSBERTa pairwise semantic similarity matrices across four global mega-suppliers (Bosch, Continental, ZF Group, Denso) during the early proprietary era (2010–2014) versus the late standardized API era (2020–2024). Panel (C) illustrates architectural layer specialization across the decoupled cyber-physical hierarchy.

4.2 Empirical Proof of Supplier Homogenization and Commoditization

Figure 5 presents empirical evidence of this vertical boundary reconfiguration:

1. **Upward Algorithmic Intrusion by OEMs:** As shown in Panel (A), the OEM share of patent filings in Vehicle Motion Control and chassis motion planning rose monotonically from 14.2% in 2010 to **68.5%** in 2024. Automakers aggressively internalized high-level vehicle dynamic control algorithms.
2. **Algorithmic Retreat of Tier-1 Suppliers:** Over the same period, the Tier-1 supplier share in motion control software fell from 85.8% to 31.5%. Suppliers were effectively pushed out of the algorithmic decision layer, retreating into physical electromechanical actuators (e.g., brake calipers, electric motors, hydraulic valves), where their patent share remained stable above 80.8%.
3. **Inter-Supplier Semantic Convergence:** Panel (B) provides striking empirical evidence regarding the homogenization of supplier innovation. In the early era (2010–2014), when suppliers developed proprietary black-box

ECUs, the pairwise PatentSBERTa cosine similarity across Bosch, Continental, ZF, and Denso averaged **0.58**, reflecting highly differentiated technical solutions. In the late era (2020–2024), under standardized OEM VMC APIs and standard CAN-FD/Ethernet communication protocols, the cross-supplier pairwise similarity surged to **0.83**.

This statistical convergence confirms the theoretical prediction of transaction cost economics and property rights theory (Grossman & Hart, 1986; Jacobides, 2022): once interface standards are established by the architectural orchestrator (the OEM), competing component suppliers are forced to conform to identical input/output specifications. Consequently, Tier-1 suppliers experience technical commoditization, shifting from proprietary system architects to standardized hardware device providers.

5 Unified Data Infrastructure and Execution Plan

5.1 Data Infrastructure and Multi-Source Linkages

The empirical credibility of this research program rests on four tightly integrated data assets:

1. **Global Autonomous Driving & Powertrain Patent Database:** A master corpus of 10,373,018 global automotive patent records from 1990 to 2024, from which 3.208M electric powertrain, 5.842M internal combustion, and 510,177 verified autonomous driving patent families are extracted.
2. **Hand-Coded Vehicle Platform Teardown Database:** Detailed engineering coding of 25 global automakers, 152 distinct vehicle platforms, and 264 vehicle models across the 2008–2024 window. Platform classifications (Legacy ICE, Conversion EV, Native Dedicated EV) are audited against physical vehicle teardown reports (A2Mac1, Caresoft, Munro & Associates) and OEM financial disclosures.
3. **768-Dimensional PatentSBERTa Semantic Embeddings:** A continuous vector representation of 489,754 patent families with complete English claims, enabling precise measurement of semantic distance and conceptual convergence (Kelly et al., 2021; Hain et al., 2022).
4. **OEM-Supplier Dyadic Relational Database:** A mapped network of 111,351 patent documents filed by the top 10 global Tier-1 suppliers, linked to automaker supply contracts and platform sourcing announcements.

Table 4: Multi-Paper Execution Roadmap and Milestone Schedule

Project Phase	Timeline	Core Analytical Tasks and Deliverables	Target Outlet
Phase 1 Study 2 Flagship	Months 1–3	Finalize staggered DiD regressions exploiting the 20-incumbent Conversion EV window; conduct Callaway-Sant'Anna dynamic event studies; finalize English manuscript.	Research Policy (Innovation Flagship)
Phase 2 Study 1 Macro	Months 4–6	Execute full-sample macro co-evolution regressions across 3.2M clean and 5.8M dirty patents; generate high-resolution alluvial migration flows; complete manuscript.	PNAS / Nature Energy (Macro Audience)
Phase 3 Study 3 Boundary	Months 7–9	Construct OEM-supplier bilateral citation matrices; estimate supplier semantic convergence regressions; complete inter-firm vertical boundary manuscript.	Strategic Management Journal (Strategy Flagship)

6 Conclusion and Theoretical Contributions

This research program establishes a new theoretical and empirical bridge between technical architecture, knowledge recombination, and firm boundaries. By demonstrating that technical architecture acts as the physical conduit for recombinant search, we show that software-hardware decoupling is not an inherent property of digital technology, but an engineered institutional and architectural achievement. In cyber-physical systems, without

modular centralization and interface abstraction, algorithmic recombination remains constrained by physical friction.

For strategy and organization science, this program provides direct empirical evidence that architectural control enables firms to redefine industry division of labor, internalize core algorithmic rents, and commoditize upstream suppliers. For managers and policymakers, our findings caution against treating digital and physical transitions as independent domains, demonstrating that leadership in artificial intelligence increasingly requires mastery of the physical architectures through which algorithms touch the physical world.

References

Acemoğlu, D., Aghion, P., Barrage, L., & Hémous, D. (2023). Climate change, directed innovation, and energy transition: The long-run consequences of the shale gas revolution. *National Bureau of Economic Research Working Paper No. 31657*.

Brunner, S., Reifinger, M., & Steinbach, T. (2017). Automotive E/E architectures and time-sensitive networking: A quantitative comparison. *IEEE Transactions on Intelligent Vehicles*, 2(4), 268–281.

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3), 414–427.

Di Natale, M., Zeng, H., Giusto, P., & Ghosal, A. (2012). *Understanding and Using the Controller Area Network Communication Protocol: Theory and Practice*. Springer.

IEEE Standards Association. (2015). *IEEE 802.1Qbv: Enhancements for Scheduled Traffic (Time-Sensitive Networking)*. IEEE.

MacKinnon, J. G., & Webb, M. D. (2018). The wild bootstrap for few (treated) clusters. *The Econometrics Journal*, 21(2), 114–135.

Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *The Review of Economic Studies*, 90(5), 2555–2591.

Robert Bosch GmbH. (2018). *Bosch Automotive Handbook* (10th ed.). Wiley.

SAE International. (2018). *SAE J1939: Recommended Practice for a Serial Control and Communications Heavy Duty Vehicle Network*. SAE International.

Saku, N., et al. (2023). PatentSBERTa: A deep learning language model for patent analysis. *World Patent Information*, 73, 102193.

Aghion, P., Dechezlepretre, A., Hemous, D., Martin, R., & Van Reenen, J. (2016). Carbon taxes, path dependency, and directed technical change: Evidence from the auto industry. *Journal of Political Economy*, 124(1), 1–51.

Brusoni, S., Prencipe, A., & Pavitt, K. (2001). Knowledge specialization, organizational coupling, and the boundaries of the firm: Why do firms know more than they make? *Administrative Science Quarterly*, 46(4), 597–621.

Ethiraj, S. K., & Levinthal, D. (2004). Modularity and innovation in complex systems. *Management Science*, 50(2), 159–173.

Ulrich, K. (1995). The role of product architecture in the manufacturing firm. *Research Policy*, 24(3), 419–440.

Baldwin, C. Y., & Clark, K. B. (2000). *Design Rules: The Power of Modularity*. MIT Press.

Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230.

Cicerone, G., Faggian, A., & Montresor, S. (2022). Regional artificial intelligence and the geography of environmental technologies: Does local AI knowledge help regional green-tech specialization? *Regional Studies*, 57(10), 2004–2018.

Coase, R. H. (1937). The nature of the firm. *Economica*, 4(16), 386–405.

Colfer, L. J., & Baldwin, C. Y. (2016). The mirroring hypothesis: Theory, evidence, and exceptions. *Industrial and Corporate Change*, 25(5), 709–738.

Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1), 117–132.

Grossman, S. J., & Hart, O. D. (1986). The costs and benefits of ownership: A theory of vertical and lateral integration. *Journal of Political Economy*, 94(4), 691–719.

Hain, D. S., Jurowetzki, R., & Buchmann, T. O. (2022). A text-embedding-based approach to measuring patent-to-patent technological similarity. *Technological Forecasting and Social Change*, 177, 121559.

Helper, S., & MacDuffie, J. P. (2023). Modularizing the automobile: Lessons from product architecture in the transition to electric, connected, and autonomous vehicles. *Management and Organization Review*.

Henderson, R. M., & Clark, K. B. (1990). Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, 35(1), 9–30.

Jacobides, M. G. (2005). Industry change through vertical disintegration: How and why markets emerge in mortgage banking. *Academy of Management Journal*, 48(3), 465–498.

Jacobides, M. G. (2022). How platforms and ecosystems reshape the boundaries of the firm: Architectural control and the commoditization of complementors. *Industrial and Corporate Change*, 31(5), 1119–1135.

Kelly, B., Papanikolaou, D., Seru, A., & Taddy, M. (2021). Measuring technological innovation over the long run. *American Economic Review: Insights*, 3(3), 303–320.

MacDuffie, J. P. (2013). Modularity-as-property, modularization-as-process, and 'modularity'-as-frame: Lessons from product architecture in the automobile utility. *Global Strategy Journal*, 3(1), 8–40.

Mazzei, J., Rughi, T., & Virgillito, M. E. (2023). Knowing brown and inventing green? Incremental and radical innovative activities in the automotive sector. *Industry and Innovation*, 31(2), 235–268.

Milgrom, P., & Roberts, J. (1990). The economics of modern manufacturing: Technology, strategy, and organization. *American Economic Review*, 80(3), 511–528.

Miric, M., Jia, N., & Huang, K. G.-L. (2022). Using supervised machine learning for large-scale classification in management research: The case for identifying artificial intelligence patents. *Strategic Management Journal*, 43(3), 491–519.

Monteverde, K., & Teece, D. J. (1982). Supplier switching costs and quasi-rents in the automobile industry. *The Bell Journal of Economics*, 13(1), 206–213.

Singh, J., & Fleming, L. (2010). Lone inventors, collaborative teams, and the discovery of breakthrough inventions. *Management Science*, 56(1), 41–56.

Verhoeven, D., Bakker, J., & Veugelers, R. (2016). Measuring technological novelty with patent data: Issues and alternatives. *Research Policy*, 45(3), 707–723.

Weitzman, M. L. (1998). Recombinant growth. *Quarterly Journal of Economics*, 113(2), 331–360.

Williamson, O. E. (1985). *The Economic Institutions of Capitalism*. Free Press.

Yayavaram, S., & Ahuja, G. (2008). Decomposability in knowledge structures and its impact on technological search: A longitudinal study. *Administrative Science Quarterly*, 53(2), 333–362.

Yoo, Y., Henfridsson, O., & Lyytinen, K. (2010). The new organizing logic of digital innovation: An agenda for information systems research. *Information Systems Research*, 21(4), 724–735.